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ORIGINAL INVESTIGATION

Improvement of Prediction of Noncontact Injury in Elite Australian Footballers With Repeated Exposure to Established High-Risk Workload Scenarios

Marcus J. Colby, Brian Dawson, Peter Peeling, Jarryd Heasman, Brent Rogalski, Michael K. Drew, and Jordan Stares

Objectives: To assess the effect of multiple high-risk-scenario (HRS) exposures on noncontact injury prediction in elite Australian footballers. ***Design:*** Retrospective cohort study. ***Methods:*** Sessional workload data (session rating of perceived exertion, global positioning system–derived distance, sprint distance, and maximum velocity) from 1 club (N = 60 players) over 3 seasons were collated; several established HRSs were also defined. Accumulated HRS sessional exposures were calculated retrospectively (previous 1–8 wk). Noncontact injury data were documented. Univariate and multivariate Poisson regression models determined injury incidence rate ratios (IRRs) while accounting for moderating effects (preseason workload volume and playing experience). Model performance was evaluated using receiver operating characteristics (area under curve). ***Results:*** Very low (0–8 sessions: IRR = 5.76; 95% confidence interval [CI], 1.69–19.66) and very high (>15 sessions: IRR = 4.70; 95% CI, 1.49–14.87) exposures to >85% of an individual's maximal velocity over the previous 8 wk were associated with greater injury risk compared with moderate exposures (11–12 sessions) and displayed the best model performance (area under curve = 0.64). A single session corresponding to a very low chronic load condition over the previous week for all workload variables was associated with increased injury risk, with sprint distance (IRR = 3.25; 95% CI, 1.95–5.40) providing the most accurate prediction model (area under curve = 0.63). ***Conclusions:*** Minimal exposure to high-velocity efforts (maximum speed exposure and sprint volume) was associated with the greatest injury risk. Being underloaded may be a mediator for noncontact injury in elite Australian football. Preseason workload and playing experience were not moderators of this effect.

Keywords: injury risk, area under curve, high velocity, team sports

Australian football (AF) is a physical game involving large running volumes, rapid directional changes, and high-velocity running efforts. Minimizing injury risk is a priority for sports medicine/science staff as injuries have a detrimental impact on team and individual success.[1] An increased understanding of the relationship between workloads and injury has been recently established.[2–15] However, these studies have only evaluated a single exposure to high injury risk situations. In practice, athletes are repeatedly exposed to these situations as components of their training and competition schedules. Preliminary research has found a protective effect with the exposure to maximum velocity efforts in Gaelic football players,[10] suggesting both overexposure and underexposure may be associated with increased injury risk. For the practitioner, these high injury risk scenarios (HRSs) present challenges to the integrated high-performance team[16,17] charged with managing the workload of athletes. Furthermore, factors such as aerobic fitness,[11] accumulated preseason workload,[4,15] playing experience,[7,18] and previous injury[19] may also moderate the injury risk associated with these scenarios.

Colby, Dawson, Peeling, and Stares are with the School of Sport Science, Exercise and Health, University of Western Australia, Crawley, Australia. Colby, Dawson, Heasman, Rogalski, and Stares are with West Coast Eagles Football Club, Subiaco, Australia. Peeling is with the Western Australian Inst of Sport, Mount Claremont, Australia. Drew is with the Australian Inst of Sport, Canberra, Australia; the Australian Centre for Research into Injury in Sport and Its Prevention (ACRISP), Federation University Australia, Ballarat, Australia; and the University of Canberra Research Inst for Sport and Exercise (UCRISE), University of Canberra, Bruce, Australia. Colby (marcus.colby@research.uwa.edu.au) is corresponding author.

A consideration for sports medicine/science staff is determining the cost–benefit ratio[20] of injury risk decisions.[16,17] Here, the outcomes of the overall high-performance system must be judged against subsystems, such as coaching needs and athlete health.[16] There is now level 1 evidence[1] to suggest that sacrificing an individual athlete's or the team's squad availability for matches will lower performance outcomes; therefore, each decision must equally account for performance and health consequences. Predictive qualities (sensitivity, specificity, and receiver operating characteristics), in addition to relative and absolute risks associated with the data, may assist in this shared decision-making process.[21] To date, only one study in AF has reported on subsequent week absolute injury risk changes (compared with baseline) for common workload metrics.[13] Here, an exposure to a high acute sprint load increased injury probability by 0.5% over the baseline exposure risk (pretest probability). However, as highlighted recently,[22] the minimal important difference for these absolute risks before targeted (indicated) prevention occurs is subject to the views of players, coaches, and/or practitioners and is context specific.

To date, most research has assessed injury risk as a "one-off" exposure at a given time point (ie, start of a week[5]) or latent period.[14] Yet, in practice, athletes may be repeatedly exposed to these potential high-risk injury situations. Therefore, this study seeks to inform practitioners in 3 ways: (1) to detail how multiple exposures to established HRSs increase injury risk, (2) to determine if a player's preseason workload exposure and playing experience moderate the HRS and injury relationship, and (3) to determine the clinical utility of these results in a real-world setting. We hypothesized that greater exposure to established HRSs would be associated with increased

injury risk, and that this relationship would be moderated by preseason workload and playing experience.

Methods

Player data (n = 60; 46 players were listed in multiple seasons) from a single Australian Football League (AFL) club across 3 consecutive seasons (2014–2016) were used. In total, 7147 individual in-season sessional data points were collected. Mean (SD) of players' age, stature, and body mass was 23.3 (3.8) years, 188.9 (6.4) cm, and 88.1 (7.9) kg, respectively. For AFL system experience, 27% of players had 1–2 years, 27% had 3–6 years, and 46% had 7 years and above. Players either competed in AFL or Western AFL matches across these seasons. All players provided written consent prior to participation. Data were deidentified and extracted from the club's athlete management system (SMARTABASE; Fusion Sport, Brisbane, Australia). Human ethics approval was obtained from the University of Western Australia's review board (approval RA/4/1/5015).

Injury information was classified and collated by the club's senior physiotherapist. As per previous research,[5] only lower-body noncontact injury resulting in matches missed was included. This definition of injury is comparable to a competition, sports incapacity injury.[23,24]

Previously validated objective (global positioning system derived; total distance, sprint distance, and maximal velocity) and subjective (field on-leg session rating of perceived exertion [sRPE]) sessional workload data were collated throughout each season, as per prior research.[5] Using the evidence base of previous literature, several HRSs were defined (Table 1). Sessions where athletes were exposed to a HRS were summed for several retrospective time frames (ie, previous 1–8 wk of training) prior to every session. It should be noted that the traditional acute to chronic workload ratio (ACWR) was used (to match previous investigations within this cohort[5]); however, recent investigations have suggested caution should be exercised with this method due to mathematical coupling.[25]

A mixed-model generalized estimating equation analyzed the relationship between sessional data and injury, as these analyses can handle panel data (repeated individual measures). For injury risk (injury/no injury in any session), a Poisson log-link regression with robust error estimate and exchangeable working correlation structure (within the generalized estimating equation model) was used.[26] To determine if multiple exposures to HRSs increased or decreased noncontact injury risk, incidence rate ratios (IRRs) were calculated for above (high risk) and below (low risk) the cut point, thereby maximizing sensitivity and specificity (Youden's index). Maximum velocity counts were sorted from lowest to highest and split into pentiles for analysis to represent very low to very high ranges as quadratic relationships for this variable had been observed previously.[10] Univariate generalized estimating equation regression models for each predictor variable were determined, not accounting for other moderating covariates.[5,27] As playing experience[18] and preseason workload[4,15] have both been previously associated with injury, these variables were entered into a multivariate model (as pentiles) to determine if the risks associated with HRS exposures were moderated.[27] Adjusted IRR (adj IRR) in the multivariate models represented the risk while accounting for moderating effects of other variables.[5] All data analysis was performed in Stata 12 (Stata 12 IC; StataCorp, College Station, TX). Statistical significance occurred when an IRR 95% confidence interval did not cross 1.00. Clinical significance was determined if the IRR was greater than 3.0.

The receiver operating characteristic curves were then used to assess the accuracy of predicted probability from each model, with area under curve (AUC) comparisons to determine the best time frame undertaken using the "jack-knife method" (a nonparametric estimate for variance comparisons[28]), with Šidák correction to account for multiple comparisons. When no significant difference occurred between time frames, the most practical (shortest) time

Table 1 Quantification of High-Risk Scenarios and Supporting Literature

High-risk scenario	Calculation	Supporting literature
	Count of sessions exposed for the past "n" weeks	
Overload/nonfunctional overreaching		
ACWR spike	Very high ACWR as determined by sessions categorized in the fifth quantile	5, 8, 9, 14
Week-to-week change	Previous (2 wk ago) to current week (last 7 d) change >10%, 20%, and 30%	6, 12
Very high chronic load	Very high 4-wk chronic load as determined by sessions categorized in the fifth quantile	3, 12
Acute workload ceiling	Individual's highest 1-wk acute load for the current season	2
Chronic workload ceiling	Individual's highest 4-wk chronic load for the current season	2
Season high maximal velocity	Individual's new maximum speed for that season	Novel—"PB effect"
Underload/ill prepared for competitive demands		
ACWR trough	Very low ACWR as determined by sessions categorized in the first quantile	5
Very low chronic load	Very low 4-wk chronic load as determined by sessions categorized in the first quantile	5
Exposure to maximal velocity	Session with at least one effort >80%, 85%, 90%, and 95% of historical maximum speed to date	10
Potential moderators		
AFL system playing experience	3 quantiles to represent a developing, main squad, or veteran athlete	5, 16
Preseason workload	3 quantiles to represent low, moderate, and high total preseason workload	4, 15

Abbreviations: ACWR, acute:chronic workload ratio; AFL, Australian Football League; PB, personal best. Note: "n" = 1 to 8 weeks.

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frame in a real-world setting was presented. Furthermore, to evaluate each model's ability to fit out-of-sample data, *k*-fold cross-validation with 10 folds was utilized. For comparison, root mean square error was reported where lower values and less variability (SD) between *k* folds indicated a better fit. To determine the clinical utility of risk factors, pretest and posttest probabilities were calculated and the difference (absolute risk) presented. Injured players' data for the sessions following injury were excluded until they returned to main training.

Results

Overall, 58 noncontact lower-body injuries resulting in games missed were sustained across the 3 in-season phases and were subsequently included in the analysis. Descriptive statistics and very low/very high cut points for workload variables are presented in Supplementary Table 1 (available online).

Table 2 presents the univariate models. If no significant differences existed between the AUCs, the most practical (shortest) time frame of exposures is presented. For the high-risk scenarios (presented in Table 2), sensitivity and specificity of each exposure range can be found in Supplementary Table 2 (available online).

An increase in the positive predictive value was observed for lower preseason (global positioning system) workloads, greater playing experience, and increased injury risk; however, the model accuracy was poor (AUC = 0.54–0.59) and the IRR 95% confidence interval crossed 1.0 (see Supplementary Table 3 [available online]). These hypothesized moderators, when included in a multivariate model, did not alter the IRR associated with the HRS, nor did they significantly improve in-sample (AUC) or out-of-sample (root mean square error) model accuracy. As such, the results are not presented here.

The pretest probability of sustaining a noncontact lower-limb injury was 0.81% per day. To display the daily predictive properties of each variable, absolute risk (posttest probability − pretest probability) is presented in Table 2.

When variables were modeled on unseen data (cross-validated), root mean square error values for all variables were similar (.09 ± .02 SD), suggesting these variables had a similar ability to fit out-of-sample data. Therefore the results are not presented here.

A practical guide for optimal maximum speed exposures over a 4-week time frame is presented in Figure 1.

Discussion

To our knowledge, this is the first study to investigate multiple exposures to high-risk loading conditions and noncontact injury risk in elite AF. Prior research[3,5,12–14] has investigated injury risk associated with single exposures to high-risk loading conditions in AF; however, our data show that greater risk may be associated with multiple exposures (which commonly occurs) over varying time frames. The HRS pertaining to high velocity exposure (maximum speed exposure and sprint chronic volume) displayed the greatest relative risks and injury predictive accuracy. For these scenarios, underload (ie, a low chronic load, low maximum speed exposure) recorded the greatest association with injury risk, potentially reflecting a state of underpreparedness. For this cohort, minimal exposure to maximum velocity efforts and having a low chronic load condition may be a key mediator for noncontact injury.

A U-shaped relationship was evident for repeated sessional exposure to near maximal velocities and noncontact injury risk (Figure 1). Very low exposures identified the greatest number of injuries (49%). Previous research[10] has investigated exposure to maximum velocity efforts in Gaelic football players but assessed neither time frames greater than the previous 7 days nor predictive accuracy. Our findings suggest that exposures over longer time frames (4–8 wk) improve predictive accuracy. Targeted (indicated) injury prevention strategies in AF should include exposure to >85% of an individual's maximum velocity over 5 to 8 sessions (training or games) across a 4-week block as this velocity threshold displayed the greatest predictive accuracy (AUC = 0.60–0.64).

A repeated exposure effect for on-leg sRPE very low (<3623 AU) chronic workloads in the previous week was observed. Subjective measures of workload may be more sensitive at predicting noncontact injury when a very low load "trough" of 2 or more sessions is observed. Furthermore, multiple exposures (3 or more) to a very high (>1.37) on-leg sRPE ACWR over the previous 2 weeks was associated with greater injury risk compared with less than 3 exposures; although sensitivity (22% of injuries above threshold) and predictive accuracy were poor (AUC = 0.55). Potentially, players eliciting a very high ACWR in a session increase their chronic load to a point that does not allow for a very high ACWR in subsequent sessions, possibly explaining this lack of findings. Finally, multiple exposures (3 or more sessions) to large week-to-week changes (on-leg sRPE > 20% and distance > 30%) over the previous 2 weeks increased injury risk but also displayed poor predictive accuracy (AUC = 0.55–0.56). Despite a low sensitivity (21%–22%) to predict injury when classified as high risk, the number of "false alarms" (specificity = 87%–90%) was notably lower than other variables, suggesting practitioners should still consider acting when these scenarios occur. If a player enters a high-risk condition, such as those identified above, an "indicated"[29] (targeted) prevention approach should attempt to minimize subsequent exposures within the following 2-week block.

For objective workload measures, as little as 1 sessional exposure to a very low chronic (distance < 75,141 m and sprint < 599 m) or acute (sprint < 118 m) load condition in the past week increased injury risk. Notably, exposure to a very low sprint chronic load condition in the previous week was associated with a 3-fold increase in injury risk, identifying 50% of injuries in this cohort. As such, preventive strategies should ensure AF players attain >150 m of sprint volume per week to maintain a minimum workload for competitive demands.[2] Practitioners may also take a selective prevention approach[29] for players forecasted to fall into a very low acute or chronic load conditions in subsequent sessions (forecasted sessions may be calculated through a training drill and/or game average database). Suggested approaches may include (1) informing the player of the low output prior to training and the need for greater exertion in training drills, (2) modifying drills within training to allow greater output for the required (low) load variables, and (3) prescribing running conditioning drills to allow players to attain the required volume.

It was hypothesized here that playing experience would be a moderating factor in significantly decreasing the IRR associated with HRSs. However, as reported recently,[7] there was no evidence to suggest playing experience significantly moderated the HRSs defined here. Greater preseason volume has also been shown to reduce in-season injury risk in this cohort[4]; however, no moderating effect was observed when this was accounted for in a multivariate model. These findings may be explained by the modeling approach undertaken here. First, HRSs were defined prior to considering any interaction effects. Therefore, HRSs may vary depending on years of playing experience. Second, the "optimal"

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Table 2 Univariate Models: High-Risk Scenario Exposures and Noncontact Injury Risk

High-risk scenario	Time frame: sessions exposed	PPV, %	IRR (95% CI)	Absolute risk	AUC (95% CI)
Exposure to max speed					
>85% individual max (most significant)	Last 8 wk:				
Very low	0–8	1.32	**5.76 (1.69–19.66)**	0.55%	
Low	9	0.58	2.58 (0.69–9.64)	–0.19%	
Moderate	11–12 (ref)	0.21	1.00	–0.55%	0.64 (0.58–0.71)
High	13–15	0.66	**3.03 (1.01–9.10)**	–0.11%	
Very high	>15	1.03	**4.70 (1.49–14.87)**	0.26%	
>85% individual max (most practical)	Last 4 wk:				
Very low	0–4	1.18	2.15 (0.97–4.79)	0.41%	
Low	5	0.49	0.93 (0.33–2.59)	–0.28%	
Moderate	6 (ref)	0.52	1.00	–0.25%	0.60 (0.53–0.68)
High	7–8	0.55	1.05 (0.46–2.42)	–0.22%	
Very high	9–12	0.87	1.64 (0.62–4.31)	0.10%	
Very low 4-wk chronic load					
Sprint distance chronic < 599 m	Last 1 wk:				
Low risk	0 (ref)	0.54	1.00	–0.27%	0.63 (0.56–0.69)
High risk	≥1	1.64	**3.25 (1.95–5.40)**	0.83%	
On-leg sRPE chronic < 3623 AU	Last 1 wk:				
Low risk	<2 (ref)	0.63	1.00	–0.18%	0.59 (0.53–0.65)
High risk	≥2	1.63	2.52 (1.51–4.19)	0.82%	
Distance chronic < 75,141 m	Last 1 wk:				
Low risk	0 (ref)	0.66	1.00	–0.15%	0.57 (0.50–0.63)
High risk	≥1	1.18	1.71 (1.09–2.66)	0.37%	
Very low 1-wk acute load					
Sprint distance < 118 m	Last 1 wk:				
Low risk	0 (ref)	0.60	1.00	–0.21%	0.59 (0.52–0.65)
High risk	≥1	1.24	2.04 (1.24–3.38)	0.43%	
Very high ACWR					
On-leg sRPE > 1.37	Last 2 wk:				
Low risk	<3 (ref)	0.63	1.00	–0.08%	0.55 (0.49–0.60)
High risk	≥3	1.16	1.93 (1.13–3.31)	0.65%	
Large week-to-week change					
On-leg sRPE > 20%	Last 2 wk:				
Low risk	<4 (ref)	0.70	1.00	–0.11%	0.56 (0.51–0.62)
High risk	≥4	1.82	2.53 (1.38–4.63)	1.01%	
Distance > 30%	Last 2 wk:				
Low risk	<3 (ref)	0.72	1.00	–0.09%	0.55 (0.49–0.60)
High risk	≥3	1.38	1.85 (1.02–3.35)	0.57%	

Abbreviations: ACWR, acute:chronic workload ratio; AUC, area under curve; CI, confidence interval; IRR, incidence rate ratio; On-leg sRPE, field session rating of perceived exertion; PPV, positive predictive value (injured cases/total cases × 100); ref, reference category. Note: High-risk scenario's where IRR > 3.00 (clinical significance) appear in bold.

cut point (that maximized sensitivity and specificity) was determined for the full squad. Creating exposure cut points by playing experience and/or preseason workload may provide greater insight.

As discussed in a recent editorial,[21] it is important to differentiate between association (ie, incidence risk ratios) and prediction (ie, sensitivity, specificity, AUC) when advising coaches on the injury risk for various loading conditions. Further to these suggestions, we believe comparing HRS to baseline exposure risk (or pretest probability) of playing the sport/being on the field will give practitioners further insight into the absolute risk in applied settings. The absolute risk (or risk difference[22]) is calculated by subtracting the posttest probability (probability after being classified within a loading condition) from the pretest probability (probability of getting injured within the total sample). When interpreting absolute risks, it is important to consider the setting in which they are calculated; for example, daily/sessional, weekly, or training phase. For this study, absolute risks were unsurprisingly low as the "event" (ie, noncontact injury resulting in a match

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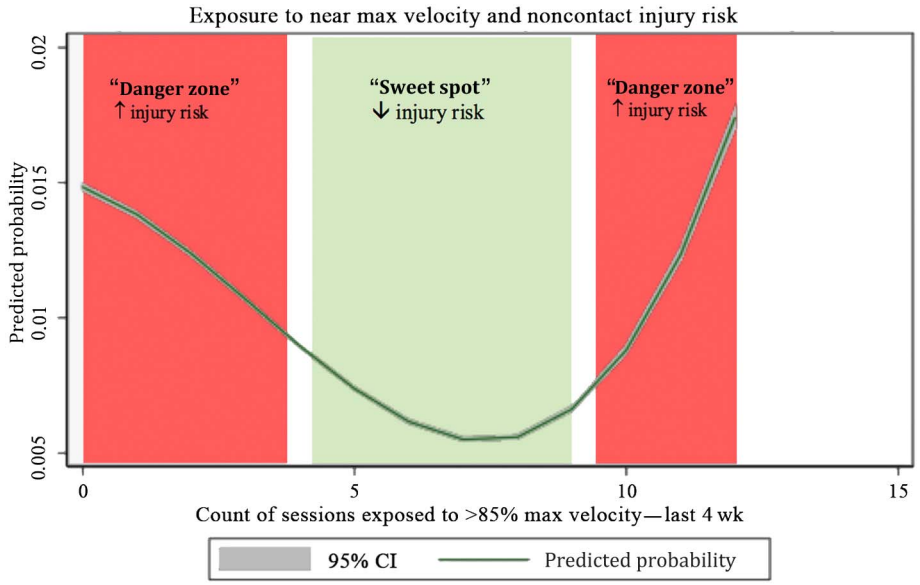

Figure 1 — Predicted injury probability for the most practical maximum speed exposure model. A U-shaped relationship for the number of sessions exposed to near maximal velocity and noncontact injury risk is evident. To minimize injury risk, athletes should aim to elicit greater than 85% of their maximum velocity in 5 to 8 sessions over a 4-week period. CI indicates confidence interval.

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missed) was rare, and data were analyzed on a sessional basis. Previous research[13] has reported a *weekly* absolute risk of 0.5% for incurring a subsequent week hamstring injury strain when exposed to >653 m of high-intensity running. Our results identified a slightly higher absolute risk (0.37%–1.01%), even when calculated on a *sessional* basis, with a very low chronic load (sprint and on-leg sRPE) and large week-to-week changes (multiple exposures >20% on-leg sRPE) displaying the greatest absolute risk. Although this study did not seek to establish the minimal important difference[22] from the absolute risks presented, it is important that coaches are informed and make injury risk decisions in a context-specific manner.

Finally, some study limitations should be acknowledged. First, a relatively low number of injuries was included (and therefore the chance of detecting an injury event was rare, being <1%) as the study focused on injuries most detrimental to performance (matches missed). Accounting for the interplay between all injuries (ie, low severity and/or high impact contact injuries) may improve injury detection as preliminary evidence suggests subsequent injuries are usually related to an initial/previous injury.[19] Second, as a retrospective study design was employed to include all available workload data up to a given session, the injury lag period (ie, subsequent week) was excluded. Future research may adopt a prospective epidemiological approach that examines the multiplicative injury risk of each subsequent exposure to a HRS and injury in a given lag period. Third, staff at the AFL club used here were aware of current literature and likely made decisions to mitigate risk (ie, training load management) when athletes were exposed to a HRS (ie, an ACWR > 1.50). By auditing injury risk actions and decisions, practitioners may further validate these metrics for injury prediction in prospective seasons. Fourth, the HRSs defined here were taken from previous literature and case studies specific to this cohort. Future research should examine the HRSs specific to a particular sport. Finally, this modeling approach examined the total sum of sessions exposed in each time frame and did not account for the distribution of these exposures. Further research should determine if consecutive or exponentially weighted exposures increase injury risk.

Conclusions

A U-shaped relationship between maximum speed exposure and increased injury risk was evident in AF. Low maximum speed exposure and low chronic workloads were identified as HRSs. Multiple exposures to HRSs may be required before noncontact injury detection is optimized. Very low daily absolute lower-limb injury risk was observed. Being underloaded may be a mediator for noncontact injury in elite AF.

Practical Implications

(1) Implementing a maximum velocity "top-up" training protocol to ensure athletes are exposed to near maximal speeds on a regular basis may reduce injury risk.

(2) Establishing a chronic load "floor" may reduce injury risk by ensuring players are well prepared for the demands of competition.

(3) Targeted injury prevention strategies in AF should involve forecasting loads to monitor players who are trending toward very low loading conditions.

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